

UNIVERSIDAD DE SONORA
Facultad Interdisciplinaria de Ingeniería
Departamento de Ingeniería Industrial



"El saber de mis hijos
hará mi grandeza"

POSGRADO EN INGENIERÍA INDUSTRIAL
MAESTRÍA EN INGENIERÍA EN INTERNET DE LAS COSAS E
INTELIGENCIA ARTIFICIAL

Clasificación de defectos en superficies planas con
sensores *Time-of-Flight*

T E S I S

PRESENTADA POR

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Desarrollada para cumplir con uno de los
requerimientos parciales para obtener
el grado de Maestro en Ingeniería

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- Alonso, M., Izaguirre, A., Andonegui, I., & Graña, M. (2020). Optical dual laser based sensor denoising for onlinemetal sheet flatness measurement using hermite interpolation [All Open Access; Gold Open Access; Green Open Access]. *Sensors (Switzerland)*, 20(18), 1-20. <https://doi.org/10.3390/s20185441>
- Alonso, M., Maestro, D., Izaguirre, A., Andonegui, I., & Graña, M. (2021). Depth data denoising in optical laser based sensors for metal sheet flatness measurement: A deep learning approach [All Open Access; Gold Open Access; Green Open Access]. *Sensors*, 21(21). <https://doi.org/10.3390/s21217024>
- Du, Y., Deng, Y., Zhou, Y., Jiao, F., Wang, B., Xu, Z., Liu, Y., Jiang, Z., Jian, S., & Guan, X. (2026). A Structure-Aware Compressed Sensing Framework for Robust Multipath Suppression in Indirect ToF Imaging. *IEEE Transactions on Instrumentation and Measurement*, 75. <https://doi.org/10.1109/TIM.2026.3652690>
- Marchisotti, D., & Zappa, E. (2022). Feasibility Study of Drone-Based 3-D Measurement of Defects in Concrete Structures [All Open Access; Green Open Access]. *IEEE Transactions on Instrumentation and Measurement*, 71. <https://doi.org/10.1109/TIM.2022.3170969>
- Molleda, J., Usamentiaga, R., García, D. F., Bulnes, F. G., & Ema, L. (2011). Shape measurement of steel strips using a laser-based three-dimensional reconstruction technique [All Open Access; Green Open Access]. *IEEE Transactions on Industry Applications*, 47(4), 1536-1544. <https://doi.org/10.1109/TIA.2011.2155019>
- Shuai, H., Chi, H., Li, Y., Wang, Y., Qian, Y., Lai, Z., & Wang, Y. (2026). 3D Laser Scanning-Based Automated Wall Surface Flatness Inspection [All Open Access, Gold Open Access]. *Buildings*, 16(14). <https://doi.org/10.3390/buildings16142911>
- Tsai, S.-Y., & Chang, J.-Y. (2021). Regression approach to a novel lateral flatness leveling system for smart manufacturing [All Open Access; Gold Open Access; Green Open Access]. *Applied Sciences (Switzerland)*, 11(14). <https://doi.org/10.3390/app11146645>
- Wang, J., Qiang, S., He, L., Chen, W., Tong, J., & Shuai, W. (2026). A LiDAR-Based Automatic Workflow for Flatness Detection of Simulated Concrete Placing Surfaces Using Multi-Frame Median Fusion [All Open Access; Gold Open Access]. *Buildings*, 16(14). <https://doi.org/10.3390/buildings16142809>

- Yang, Y., Ye, Z., Wang, Y., Liu, C., & Kan, C. (2024). Recurrence analysis of 3D point clouds for the detection of surface anomalies in advanced manufacturing [All Open Access; Hybrid Gold Open Access]. *Manufacturing Letters*, 41, 1499-1506. <https://doi.org/10.1016/j.mfglet.2024.09.176>
- Yoshida, M., Hayashi, D., De Xing, L., Yasutomi, K., Kawahito, S., Kagawa, K., & Nagahara, H. (2026). Single-Shot Efficient Depth Imaging Based on Time-Compressive CMOS Image Sensor [All Open Access; Hybrid Gold Open Access]. *IEEE Sensors Journal*, 26(7), 9493-9502. <https://doi.org/10.1109/JSEN.2026.3655349>